

Fetal movements are related to maternal characteristics: a prospective ultrasound study



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BACKGROUND: Fetal movements are important indicators of well-being that are associated with stillbirth. Yet neither a universal definition nor gold standard exists for what constitutes normal or abnormal. Current monitoring methods such as maternal perception lack diagnostic accuracy, while ultrasound provides detailed assessment in real time that is not practical outside the clinic setting. Maternal perception remains the primary method of fetal movements detection in clinical practice.

OBJECTIVES: To assess normal fetal movements patterns using real-time ultrasound and maternal perception and relating these to key pregnancy parameters. To investigate how maternal perception is predictive for ultrasound identified movements.

STUDY DESIGN: Prospective observational study in an inner-city maternity hospital in the United Kingdom, enrolling a mixed-risk, multi-ethnic cohort of 61 participants with singleton pregnancies. Participants underwent two ultrasound scans at 32–34+0 weeks and 36–38+0 weeks gestation, during which ultrasound detected fetal movements and maternal perception of fetal movements were concurrently recorded. Key demographic and pregnancy characteristics were collected and compared to ultrasound movement data sets. Sensitivity of maternal perception for ultrasound detected fetal movements was calculated.

RESULTS: Neither total ultrasound-detected fetal movements nor maternal perception were significantly associated with placental position, fetal sex, parity, estimated fetal weight centile, or umbilical artery pulsatility index.

However, increasing maternal age was associated with more maternally perceived fetal movements at 36–38 weeks ($p=0.025$, $\text{coeff}=3.67$), and a lower body mass index (BMI) with more ultrasound-detected ($p=0.048$, $\text{coeff}=-16.72$) and maternally perceived ($p=0.025$, $\text{coeff}=-3.62$) movements at 36–38 weeks. Using ultrasound detected fetal movements as the gold standard, maternal perception showed a low sensitivity, with no significant difference between gestational ages: perceived movements represented 10.0% of total at 32–34 weeks and 11.4% at 36–38 weeks.

CONCLUSIONS: Although it is widely believed that factors such as placental location, parity, and fetal sex influence fetal movements, we report that only BMI and maternal age had a significant impact on maternal perception of fetal movements. Although the number of ultrasound-detected fetal movements did not increase with maternal age, older participants perceived more fetal movements. Moreover, the association of obesity with lower ultrasound-detected and maternally perceived fetal movements suggests that BMI may independently affect fetal movements and their perception. Maternal perception has a low sensitivity, suggesting that many movements are missed. Maternal perception is therefore not a reliable diagnostic tool, and this may underlie why studies on fetal movements and perinatal outcome are universally negative.

Key words: wellbeing, in utero, perception, third trimester

Objective

To compare maternally perceived fetal movements with real-time ultrasound-detected movements in the third trimester and to evaluate the influence of maternal and pregnancy characteristics on both measures.

Study design

In this prospective observational study, we enrolled 61 participants with singleton pregnancies from a mixed-risk, multi-ethnic population at a UK tertiary maternity unit. Ultrasound assessments

were performed at 32–34 weeks ($n=61$) and 36–38 weeks' gestation ($n=56$).

During standardized 30-minute sessions, fetal movements were recorded in real time. Maternal perception was recorded concurrently using an event marker. Concordance was defined as maternal perception within 5 seconds of an ultrasound-detected movement.

Associations between movement counts and maternal age, body mass index (BMI), parity, placental location, fetal sex, estimated fetal weight, and umbilical artery pulsatility index were evaluated using multivariable linear regression. Maternal sensitivity was defined as the proportion of ultrasound-detected movements perceived maternally. Given the sample size and number of covariates, analyses were considered exploratory.

Ultrasound operators verbally identified movements in real time, and participants were not blinded to these call-outs.

Results

A total of 59 and 54 analyzable ultrasound sessions were available at 32–34 and 36–38 weeks, respectively, with corresponding maternal perception data for 56 and 51 sessions. Total ultrasound-detected and maternally perceived movements did not differ between gestational windows.

Maternal perception demonstrated low sensitivity, identifying 10.0% (interquartile range [IQR] 4.3–18.0) of ultrasound-detected movements at 32–34 weeks and 11.4% (IQR 5.1–19.3) at 36–38 weeks ($p=0.64$). These estimates reflect the strict temporal definition of

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TABLE 1

Multivariable linear regression representing total number of Ultrasound detected fetal movements at 32–34 and 36–38 weeks in respect of all parameters

Variables	Total ultrasound movements - Multivariable analysis					
	32–34 weeks			36–38 weeks		
	Coeff	95% CI	p-value	Coeff	95% CI	p-value
Age	9.52	–6.83; 25.87	0.248	7.55	–8.98; 24.07	0.363
BMI	–3.65	–20.54; 13.24	0.667	–16.72	–33.31; –0.12	0.048
Nulliparity	26.27	–146.32; 198.87	0.761	–32.20	–202.66; 138.26	0.706
Female fetal sex	–89.32	–244.32; 65.68	0.253	46.67	–109.13; 202.47	0.550
Anterior placental site	–17.08	–167.53; 139.38	0.855	–107.89	–264.24; 48.47	0.172

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concordance and inclusion of all movement types.

In adjusted analyses, higher maternal BMI was associated with fewer ultrasound-detected movements ($p=0.048$) (Table 1) and fewer maternally perceived movements ($p=0.025$) at 36–38 weeks. Increasing maternal age was associated with greater maternal perception ($p=0.025$) (table 2) but not with ultrasound-detected movements.

Placental location, parity, fetal sex, estimated fetal weight, and umbilical artery Doppler indices were not associated with either ultrasound-detected or maternally perceived movements. No maternal or pregnancy characteristic was associated with maternal sensitivity.

Conclusion

Maternal perception identified a small proportion of movements detected during ultrasound observation under the study conditions. However, this estimate is dependent on methodological definitions and may underestimate clinically meaningful perception, which reflects broader changes in fetal activity rather than detection of individual movements.

Ultrasound was used as a reference method but is not a true gold standard, as detection is influenced by operator dependency and technical limitations, particularly in individuals with higher BMI. The observed association between higher BMI and fewer ultrasound-

detected movements may therefore reflect a combination of biological and technical factors, including reduced image quality and signal attenuation, rather than a true reduction in fetal activity alone. Furthermore, the number of participants with BMI ≥ 30 was limited, and results should be interpreted with caution.

The interactive study design, including real-time verbalization of movements, represents an important limitation, as it may have altered maternal attention and reporting. The prespecified 5 second interval between ultrasound ‘call out’ and maternal perception was chosen to account for expected delays in maternal recognition

TABLE 2

Multivariable linear regression representing maternally perceived movements <5 seconds from an Ultrasound detected fetal movement at 32–34 and 36–38 weeks in respect of all parameters

Variables	Maternally perceived movements < 5 sec - Multivariable analysis					
	32–34 weeks			36–38 weeks		
	Coeff	95% CI	p-value	Coeff	95% CI	p-value
Age	5.28	–0.06; 10.61	0.052	3.67	0.48; 6.86	0.025
BMI	–1.72	–7.30; 3.86	0.539	–3.62	–6.76; 0.47	0.025
Nulliparity	1.69	–54.73; 58.11	0.952	1.89	–30.32; 34.11	0.906
Female fetal sex	–20.04	–71.86; 31.77	0.441	7.21	–23.00; 37.41	0.633
Anterior placental site	–11.79	–63.07; 39.49	0.646	–3.01	–32.80; 26.79	0.840

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and response. Although it is not a validated threshold and may influence sensitivity estimates intervals of <2, 2, 3 and 6 seconds have been reported.^{1–3} In addition, the relatively small sample size and use of multivariable models limit generalizability and raise the possibility of overfitting.

Despite these limitations, the findings challenge commonly held assumptions regarding factors influencing fetal movement perception, including placental location and parity. Contrary to popular perception, anterior placental position is not associated with fewer maternally perceived or ultrasound detected movements. However the discrepancy between perceived and observed movements highlights the complexity of interpreting fetal movement patterns in clinical care.

Rather than indicating that maternal perception lacks clinical value, these data suggest that subjective reporting alone does not fully capture fetal activity. This finding supports ongoing efforts to develop objective and scalable monitoring approaches, including wearable sensor-based technologies, which may complement existing clinical assessments.⁴ Further studies incorporating clinical outcomes are required to determine the prognostic significance of objectively measured fetal movements. ■

CRedit authorship contribution statement

Sohini Patel: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Isabella Abati:** Writing – original draft, Methodology, Investigation,

Formal analysis, Data curation. **Giulia Zamagni:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Tricia Adjei:** Software, Resources, Methodology, Investigation, Data curation. **Saksham Dhawan:** Software, Resources, Methodology, Investigation, Data curation, Conceptualization. **Sebastian Mancero-Castillo:** Visualization, Software, Resources, Methodology, Investigation, Data curation. **Zhenhua Yu:** Software, Resources, Methodology, Data curation. **Thomas Liney:** Formal analysis, Data curation. **Susanna Inzoli:** Writing – original draft, Project administration, Investigation, Data curation. **Antonio De Sanctis:** Software, Resources, Investigation, Data curation. **Liam Corette:** Software, Resources, Investigation, Data curation. **Lorenzo Monasta:** Supervision, Data curation, Conceptualization. **Ravi Vaidyanathan:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Christoph Lees:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Data availability

The dataset for this original research is available on request to the authors.

Ethics

All procedures were performed in compliance with relevant laws and institutional guidelines and have been approved by the appropriate institutional committee(s).

The study has been approved by the London Central Research Ethics Committee (19th October 2023) (ref 23/PR/0859) & Health Regulation Authority approval 19th October 2023. Imperial Healthcare NHS Trust capacity and capability confirmation (CCC) and Sponsor 'green light' 31st October 2023 and study commenced 31st October 2023.

The privacy rights of human participants have been observed and informed consent was obtained.

Supplementary materials

Supplementary material associated with this article can be found in the online version at [doi:10.1016/j.ajogmf.2026.102043](https://doi.org/10.1016/j.ajogmf.2026.102043).

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